

Lecture 4

Probability and Random Variables

Part 1.

Probability

Probability and Uncertainty

- ❑ Probability provides a formal language to describe uncertainty.
- ❑ In business and finance, many quantities are not known in advance: future demand, investment returns, project revenues, default events, or customer behaviour.
- ❑ Probability allows us to assign numerical weights to possible outcomes and to reason systematically about risk and uncertainty.

| Outcomes and Sample Space

An **outcome** is one possible result of an uncertain situation.

The **sample space** is the set of all possible outcomes: $S = \{s_1, s_2, \dots, s_n\}$

The outcomes in a sample space must be:

- ❑ **mutually exclusive**: only one outcome can occur;
- ❑ **collectively exhaustive**: one of the outcomes must occur.

In very simple cases, the sample space is obvious. A coin has two sides, and a standard die has six faces:

$$S = \{1, 2, 3, 4, 5, 6\}$$

In business applications, defining the sample space is often already part of the analysis. For example, for a product launch:

$$S = \{Failure, Moderate\ success, High\ success\}$$

The sample space is therefore a modelling choice. It depends on the problem, the available information, and the level of detail required.

Assigning Probabilities

Once the sample space has been defined, we assign probabilities to the possible outcomes

A probability is a number between 0 and 1: $0 \leq P(\text{outcome}) \leq 1$

The probabilities of all possible outcomes must sum to 1:

$$\sum_i p_i = 1$$

For a fair die:

$$P(1) = P(2) = \dots = P(6) = \frac{1}{6}$$

In business applications, probabilities may come from different sources:

- ❑ a theoretical model,
- ❑ historical data and observed frequencies,
- ❑ subjective assessments based on expert judgment. .

The same set of outcomes may therefore lead to different analyses if different probabilities are assigned

Events

An **event** is a subset of the sample space.

If the sample space is: $S = \{s_1, s_2, \dots, s_n\}$, then an event A is a **collection of possible outcomes**:
$$A \subseteq S$$

For example, suppose a company evaluates a product launch with the following sample space:

$$S = \{\text{Failure}, \text{Moderate success}, \text{High success}\}$$

The event “the launch is successful” can be defined as:

$$A = \{\text{Moderate success}, \text{High success}\}$$

The probability of this event is obtained by adding the probabilities of the outcomes that belong to it: $P(A) = P(\text{Moderate success}) + P(\text{High success})$

If: $P(\text{Moderate success}) = 0.50$, and: $P(\text{High success}) = 0.20$,
then $\Rightarrow$ **$P(A) = 0.70$**

Events Can Overlap

Outcomes in the sample space must be **mutually exclusive**: only one outcome can occur. Events, however, are collections of outcomes. They can overlap.

For example, in a product launch analysis, we may define two events:

$$A = \{the\ launch\ is\ commercially\ successful\}$$
$$B = \{the\ launch\ generates\ high\ strategic\ value\}$$

These events are not necessarily mutually exclusive. A product launch may be both commercially successful and strategically valuable.

The overlap is written as:

$$A \cap B$$

and represents the event where both conditions hold.

The union is written as:

$$A \cup B$$

and represents the event where at least one of the two conditions holds.

When events overlap, their probabilities cannot simply be added:

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

The term $P(A \cap B)$ removes the overlap that would otherwise be counted twice.

Example: Business Scenarios

Consider a company evaluating a new product launch.

A simple sample space could be: **$S=\{\text{Failure, Moderate success, High success}\}$**

One analyst may assign the probabilities:

.

Scenario	Probability
Failure	0.30
Moderate success	0.50
High success	0.20

These probabilities reflect the analyst's view of uncertainty.

They could be based on market research, previous launches, statistical models, or managerial judgment.

Complement of an Event

If A is an event, its **complement** is the event that A does not occur.

$$A^c = \text{not } A$$

The probabilities satisfy:

$$P(A^c) = 1 - P(A)$$

Examples:

- If A is “the customer buys the product”, then A^c is “the customer does not buy the product”.
- if A is “*The return is between -2% and 3%*”, Then A^c is “*The return is less than -2% or more than 3%*”

Mutually Exclusive Events

Two events are **mutually exclusive** if they cannot happen at the same time.

$$A \cap B = \emptyset$$

Therefore:

$$P(A \cap B) = 0$$

For mutually exclusive events:

$$P(A \cup B) = P(A) + P(B)$$

Mutually Exclusive Events - Example

Let:

□ A = “the return is less than -2%”,

$$A = \{r < -2\%\}$$

□ B = “the return is more than 3%”.

$$B = \{r > 3\%\}$$

□ C = “the return is positive”.

$$C = \{r > 0\}$$

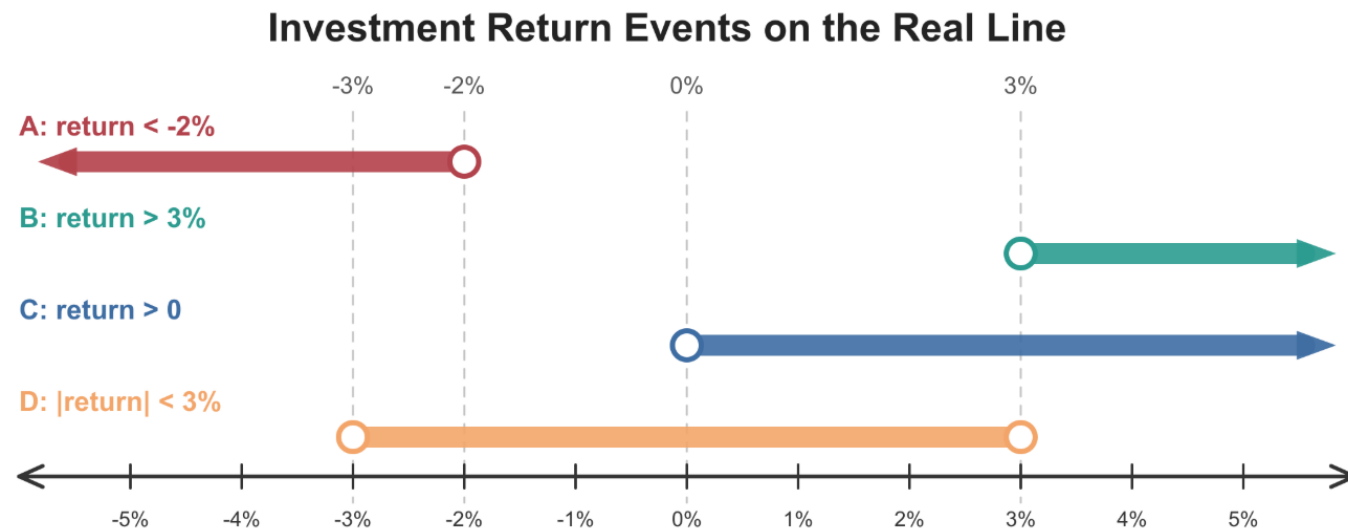
□ D = “The absolute value of the return is less than 3 %”

$$D = \{|r| < 3\%\} = \{-3\% < r < 3\%\}$$

For each pair of events, determine whether the two events are mutually exclusive.

Mutually Exclusive Events – Example – Solution

Pair	Mutually exclusive?	Why
A and B	Yes	R cannot be below -2% and above 3% .
A and C	Yes	A is negative, while C is positive.
B and C	No	$B \subset C$: if $R > 3\%$, then $R > 0$.
A and D	No	They overlap when $-3\% < R < -2\%$.
B and D	Yes	B requires $R > 3\%$, while D requires $-3\% < R < 3\%$.
C and D	No	They overlap when $0 < R < 3\%$.



Open circles mean the boundary value is not included.

Conditional probability

Conditional probability measures the probability of an event given that another event has already occurred.

We write: $P(A | B)$, and read it as: **probability of A given B**

If $P(B) > 0$, then

$$P(A | B) = \frac{P(A \cap B)}{P(B)}$$

Interpretation:

When we condition on B , we restrict attention to the cases where B occurs. Among those cases, we ask how often A also occurs.

Example conditional probability

The table shows the region and sector breakdown of a portfolio. Each cell gives the number of firms. There are 42 firms in total. Let **E** = Europe, **US** = US, **T** = Telecom, **B** = Banking

I pick a company at random.

1. What is the probability that this company belongs to the banking sector?
2. If the company is known to be from US, what is the probability that this company belongs to the banking sector?

market / sector	Telecom	Banking	Total
Europe	5	12	17
US	18	7	25
Total	23	19	42

Example conditional probability - Solution

Let $A = \{\text{company in Banking sector}\}$ and $B = \{\text{US company}\}$. By looking at the table we can use a frequentist approach to probability. (so we implicitly assume that all companies in the table are equally likely to be selected. Then:

Question 1

Probability that this company belongs to the banking sector? $\Rightarrow P(A) = 19/42 \approx 45\%$

Question 2.

The probability that this company belongs to the banking sector given that it is from the US can be rewritten as

$$P(\text{Banking} | \text{US}) = P(A | B) = \frac{P(A \cap B)}{P(B)} = \frac{7/42}{25/42} = \frac{7}{25} \approx 28\%$$

market / sector	Telecom	Banking	Total
Europe	5	12	17
US	18	7	25
Total	23	19	42

This example shows that knowing the company's country changes the probability we assign to its sector. This leads naturally to the concept of independence, which we will introduce next.

Independent Events

Two events A and B are **independent** if knowing that one occurred does not change the probability of the other.

$$P(A | B) = P(A)$$

Equivalently:

$$P(A \cap B) = P(A)P(B)$$

In the previous example:

A = “the company is based in the US”

B = “the company belongs to the banking sector”

The two events were **not independent**.

Knowing that the company was based in the US changed the probability that it belonged to the banking sector.

In business and finance, independence is often a strong assumption. It may be difficult to test, and even harder to prove, because events can be connected through hidden factors such as market conditions, sector exposure, or macroeconomic news.